
What a Benign-Trained Anomaly Detector Can and Cannot See: A Real-Data Study of the Masquerade–Traitor Boundary on TWOS

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Abstract

Benign-trained anomaly detectors are evaluated by how well their scores rank attacks, but a score says nothing about *which* attacks a detector can see at all. We study this directly on TWOS, a dataset of 24 real users with genuine self-reported personality inventories and two distinct kinds of staged insider activity—*masquerades*, in which an attacker operates a victim’s live session, and *traitors*, in which a legitimate user turns malicious. Adapting a general-purpose language model to the benign population and auditing it with sparse features and causal interventions, we find a clean behavioral detector (top causal features carry essentially no profile mass despite real psychometrics being present in every input), replicating on real data the behavioral pole of a profile-versus-behavior dissociation established on synthetic benchmarks. The detector reaches a mean within-user ROC-AUC of 0.783 over 16 attacked users from mouse, keystroke, and host-event telemetry alone. But its central finding is a *boundary*: masquerade windows are detected far better than traitor windows (0.780 vs. 0.256 pool ROC-AUC), and—crucially—this gap reproduces across four unrelated detector families (Isolation Forest, one-class SVM, PCA reconstruction, and the language model), so it is a property of the *observation space*, not of any one model: traitors are behaviorally *calmer* than their own baselines, exfiltrating out of the telemetry channel. A case study on a near-zero-activity user shows the converse—that detectability is a property of the serialized field set: enriching the serialization with a host-event channel flips that user’s within-user ROC-AUC from 0.005 (inverted) to 0.929. We argue that reporting what a detector *structurally cannot* see belongs alongside its detection numbers.

1 Introduction

An insider-threat detector trained only on benign activity is usually reported by a single kind of number: an ROC-AUC or a precision at some alert budget, summarizing how well its anomaly score separates known attacks from normal behavior. Such numbers answer “how well does it rank the attacks we gave it?” They are silent on two prior questions that matter at least as much in deployment: *what evidence* is the score actually built on, and *which kinds of attack are even visible* in the data the detector consumes.

In companion work we developed a mechanistic audit that answers the first question for adaptation-based sequence detectors, and used it to expose a dissociation: on one synthetic benchmark a benign-trained language model scores anomalies through user-identity and profile tokens rather than behavior, while on another it uses genuine behavioral evidence. That study reserved a real-data benchmark, TWOS (Harilal et al., 2017), for a compact confirmation that the behavioral pole is not a synthetic-data artifact. Here we return to TWOS as the primary object of study, because it uniquely supports the *second* question. TWOS

contains two operationally different insider scenarios under one schema—masquerades and traitors—and it pairs high-frequency behavioral telemetry with each user’s real IPIP-50 Big-Five inventory. This lets us ask not just how well a benign-trained detector scores, but where its blind spots are and why.

Our contributions are:

1. **A real-data audit with real psychometrics.** Adapting Qwen2.5-3B to the benign TWOS population and running the full sparse-feature-plus-causal audit, we find the top causally implicated features are essentially 100% behavioral at both audited layers and both adapter seeds—despite a real personality PSY line in every window. On real data the detector does not take the identity shortcut, exactly in the population regime (50% attack prevalence) where identity is a useless separator.
2. **A model-class-independent masquerade–traitor boundary.** The language model detects masquerades far better than traitors (0.780 vs. 0.256), and this dichotomy holds across Isolation Forest, one-class SVM, and PCA reconstruction on the same features. Because four unrelated detector families agree, the boundary is a statement about the telemetry, not the model: traitor activity is behaviorally quiet.
3. **A case study localizing detectability to the field set.** A near-zero-baseline user whose masquerader was the most hyperactive attacker in the dataset is scored *inverted* (ROC-AUC 0.005) by a two-source serialization, and correctly (0.929) once a host-event channel is added—a concrete demonstration that what a detector can see is a property of what the serialization exposes.
4. **A causal signature that transfers.** The estimand asymmetry observed on synthetic data—ablation-dependence is a robust causal signature of behavioral feature families while patch-repair is configuration-sensitive —reproduces a fourth time here, on real data.

We take from this that a detector’s report should include not only its scores but an explicit account of what its observation space renders invisible; on TWOS, that is precisely the traitor.

2 The TWOS benchmark and serialization

TWOS (Harilal et al., 2017) records 24 real users over five working days across six telemetry sources (keystroke, mouse, host/event-viewer, network, email, and logon), together with each user’s real self-administered IPIP-50 Big-Five personality inventory. Ground truth comprises 12 staged *masquerade* sessions (90 minutes each, in which an attacker uses a victim’s unlocked machine) and 5 staged *traitor* sessions (120 minutes each, in which a legitimate user acts maliciously). This is the combination we need: a genuine psychometric axis, two attack modalities, and a prevalence (12 of 24 users involved in masquerades) that is the opposite extreme from the small distinctive minority found in the synthetic benchmark where profile capture occurred.

We serialize the telemetry into the same DAY/PSY/SES text schema used in the companion audit, at a 10-minute window grain: a DAY line with team and machine context, a PSY line carrying the real scored OCEAN values, and SES lines summarizing per-window mouse, keystroke, and host-event activity. This yields 18,048 windows, 138 positive windows (101 masquerade, 37 traitor), and a 12-user benign training pool. We adapt Qwen2.5-3B on the benign windows only (QLoRA, three epochs, two adapter seeds), score windows by adapted negative log-likelihood, and run the identical downstream stack (token-delta extraction at layers 12/18/24; benign-only top- k sparse autoencoder; full-pool feature selection; token attribution; causal patching and necessity ablation). No component of the pipeline is TWOS-specific; the serialization is the only new code.

3 The detector is clean and behavioral

Attribution. On real data with real personality scores present in every window, the audit certifies a behavioral detector. The top-five causally selected features carry essentially all of their attribution mass on SES (behavioral) tokens and none on PSY/DAY (profile) tokens: at layer 24, seed 42, the five features have

Table 1: Window ROC-AUC by attack modality on TWOS, across four detector families trained benign-only on the same serialized features. Every family detects masquerades far better than traitors; the language model is not an outlier. “Within-user” averages per-user ROC with identity held fixed (victim’s own benign vs. attack windows).

Detector	All	Masquerade-only	Traitor-only	Mean within-user
Language model (ours)	0.639	0.780	0.256	0.783
Isolation Forest	0.714	0.839	0.374	0.715
One-class SVM	0.607	0.763	0.183	0.626
PCA reconstruction	0.790	0.867	0.582	0.795

profile mass $[0, 0, 0, 0, 0]$ and behavioral mass ≥ 0.987 ; the worst single value across both audited layers and both adapter seeds is 0.983 behavioral. This is the fifth clean replication counting the smaller two-source corpus, and it holds whether or not the enriched host-event channel is included.

This is the outcome the population-structure account predicts. With 12 of 24 users malicious, user identity carries almost no information about which windows are attacks; a next-token detector therefore has no cheap identity-shortcut to take, and the behavioral session evidence is what survives selection. The synthetic benchmark that exhibited profile capture sat at the opposite corner (four highly distinctive malicious users among thousands); TWOS confirms the behavioral corner with genuine, rather than simulated, psychometrics.

Causal support. Team-matched, same-user-excluded interventions on the seed-42 layer-24 top features (138 receivers) give a positive necessity effect (+0.0019) and an absent patch-repair effect (−0.0018). The direction reproduces the estimand asymmetry documented three times on CERT: ablation-dependence is the robust causal signature of behavioral families, while patch-repair is configuration-sensitive—here plausibly because a masquerade anomaly is another *human’s* behavior, so repairing toward a benign teammate need not reduce anomaly the way a synthetic profile swap does. The signature is weak in absolute terms (small pool, thin per-team donor sets) and we report it as directional corroboration, not as a headline effect.

4 The masquerade–traitor boundary is a property of the data

The detector’s most consequential result is not a level but a *gap*. Table 1 scores every detector on the same 10-minute windows, split by attack modality.

Two things are visible. First, masquerade detection is strong and traitor detection is at or below chance for *every* detector: the language model (0.780 vs. 0.256), Isolation Forest (0.839 vs. 0.374), one-class SVM (0.763 vs. 0.183), and PCA reconstruction (0.867 vs. 0.582) all show the same ordering with a large margin. Because these four families make completely different modeling assumptions (density, tree-isolation, margin, linear reconstruction, and sequence likelihood), their agreement cannot be a modeling artifact. The boundary is a property of the *observation space*.

Second, the reason is interpretable. A masquerade is a *foreign* operator on a victim’s machine, so the victim’s window suddenly looks unlike the victim—high, out-of-distribution mouse and keystroke activity that every anomaly model flags. A traitor is the *legitimate* user, and their malicious phase tends to be behaviorally *calmer* than their own baseline (disengagement from normal work) while the malicious act itself—data staged for exfiltration, sent over email—occurs largely *outside* the mouse/keystroke/host-event channels we serialize. The telemetry that makes masquerades loud makes traitors quiet. This is not a failure that a better model fixes; it is a limit of what the field set can express, and any detector restricted to this observation space inherits it.

The practical consequence is a reporting norm. A benign-trained detector on this telemetry should not be summarized by its pooled ROC-AUC, which averages a detectable modality with an undetectable one; it should be reported per modality, with the traitor result labeled as an observation-space blind spot rather

than a tuning target. The remedy is a channel, not a loss function: detecting traitors requires serializing the exfiltration surface (email/network content), not a better anomaly score over mouse and keyboard.

5 Case study: detectability follows the field set

If the traitor boundary shows what the observation space *hides*, one user shows the converse—that enriching the field set changes what a detector can see. User 21 has a near-empty benign baseline: a median of one mouse event per 10-minute window (all quartiles at 1.0; mean 91) and roughly 25 keystrokes—this user barely touched the monitored machine. Their masquerader, by contrast, was the single most active attacker in TWOS (23,552 mouse events per window, against 1–2,216 for every other user).

Under a two-source serialization (mouse and keystroke only), this user is scored *inverted*: within-user ROC-AUC 0.005, i.e. the hyperactive attack windows are ranked *less* anomalous than the near-empty benign ones. The mechanism is degeneracy: near-empty benign windows produce pathological likelihoods that out-rank genuine attack windows, so the raw score inverts. Adding a host-event channel to the serialization—which gives the benign windows non-degenerate content and the attack windows a second axis of evidence—eliminates the inversion entirely: User 21’s within-user ROC-AUC becomes 0.929.

The lesson generalizes the paper’s thesis: the pathological case was not a model deficiency but a serialization deficiency, and it was fixed by exposing a field, not by retraining a better scorer. Detectability is a property of what the serialization renders observable. Reporting a detector therefore means reporting its field set, because two serializations of the same underlying telemetry can place the same attack on opposite sides of chance.

6 Discussion

Read together, the three results make a single argument about how benign-trained anomaly detectors should be evaluated and reported. The clean behavioral audit shows that *what* a detector scores on is not settled by its score and must be checked. The masquerade–traitor boundary shows that *which* attacks a detector can see is often not a property of the detector at all, but of the telemetry it is given—so a good pooled number can hide a structurally invisible attack class. The User 21 case shows the same coin’s other face: the field set can be changed, and doing so, not tuning the model, is what moves an attack across the detectability boundary.

None of these are visible at the level of a leaderboard. A detector that scores 0.78 pooled on TWOS looks respectable; the audit and the per-modality split reveal that it is an excellent masquerade detector, a non-detector of traitors, and a scorer whose behavior on sparse users depends entirely on which channels were serialized. We suggest that reports of benign-trained insider-threat detectors carry three artifacts alongside their scores: an attribution of what evidence the score rests on, a per-modality breakdown that names any observation-space blind spot, and the field set under which the numbers hold.

Limitations. TWOS is small (24 users, five days, 37 traitor windows), so all numbers here are pilot-scale and single-corpus; the causal interventions in particular rest on thin donor pools and are reported as directional. The traitor account is a structural hypothesis supported by the cross-model agreement and by the calmer-than-baseline activity profile, not by instrumented exfiltration ground truth; confirming it would require serializing the email and network channels and re-running the split. We regard the boundary statement—not the individual ROC values—as the transferable claim.

References

Athul Harilal, Flavio Toffalini, John Castellanos, Juan Guarnizo, Ivan Homoliak, and Martín Ochoa. Twos: A dataset of malicious insider threat behavior based on a gamified competition. In *Proceedings of the 2017 International Workshop on Managing Insider Security Threats*, pp. 45–56, 2017.